Handling Uncertainty

Logic and Uncertainty

Major problem with logical-agent approaches:

Agents almost never have access to the whole truth about their environments

- Very often, even in simple worlds, there are important questions for which there is no yes/no answer
- In that case, an agent must reason under uncertainty
- Uncertainty also arises because of an agent's incomplete or incorrect understanding of its environment

Uncertainty

Let action A_t =“leave for airport t minutes before flight” Will A_t get me there on time?

Problems

- partial observability (road state, other drivers’ plans, etc.)
- noisy sensors (unreliable traffic reports)
- uncertainty in action outcomes (flat tire, etc.)
- immense complexity of modeling and predicting traffic

Uncertainty

Let action A_t =“leave for airport t minutes before flight” Will A_t

get me there on time?

A purely logical approach either

1. risks falsehood (“A_{25} will get me there on time”), or
2. leads to conclusions that are too weak for decision making(“A_{25} will get me there on time if there’s no accident on the way, it doesn’t rain, my tires remain intact, ...”)

(A_{1440} might reasonably be said to get me there on time but I’d have to stay overnight in the airport …)

Reasoning under Uncertainty

A rational agent is one that makes rational decisions — in order to maximize its performance measure)

A rational decision depends on

- the relative importance of various goals
- the likelihood they will be achieved
- the degree to which they will be achieved

Handling Uncertain Knowledge

Reasons FOL-based approaches fail to cope with domains like, for instance, medical diagnosis:

- Laziness: too much work to write complete axioms, or too hard to work with the enormous sentences that result
- Theoretical Ignorance: The available knowledge of the domain is incomplete
- Practical Ignorance: The theoretical knowledge of the domain is complete but some evidential facts are missing

Degrees of Belief

In several real-world domains the agent's knowledge can only provide a degree of belief in the relevant sentences

The agent cannot say whether a sentence is true, but only that is true $x\%$ of the times

The main tool for handling degrees of belief is Probability Theory

The use of probability summarizes the uncertainty that stems from our laziness or ignorance about the domain

Probability Theory

Probability Theory makes the same ontological commitments as First-order Logic:

Every sentence ϕ is either true or false

The degree of belief that ϕ is true is a number P between 0 and 1

$P(\phi) = 1$	$\dashrightarrow$	ϕ is certainly true
$P(\phi) = 0$	$\dashrightarrow$	ϕ is certainly not true
$P(\phi) = 0.65$	$\dashrightarrow$	ϕ is true with a 65% chance

Probability of Facts

Let A be a propositional variable, a symbol denoting a proposition that is either true or false

$P(A)$ denotes the probability that A is true in the absence of any other information

Similarly,

$P(\neg A)$	=	probability that A is false
$P(A \wedge B)$	=	probability that both A and B are true
$P(A \vee B)$	=	probability that either A or B (or both) are true

Prior/Unconditional Probability

$P(A)$ is the unconditional (or prior) probability of fact A

An agent can use the unconditional probability of A to reason about A in the absence of further information

If further evidence B becomes available, the agent must use the conditional (or posterior) probability:

$$P(A \mid B)$$

the probability of A given that (all) the agent knows (is) B

Note: $P(A)$ can be thought as the conditional probability of A with respect to the empty evidence: $P(A) = P(A \mid)$

Posterior/Conditional Probabilities

Conditional probabilities are defined in terms of unconditional ones

Whenever $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

Random Variable

A random variable is a variable ranging over a certain domain of values

It is

- discrete if it ranges over a discrete (that is, countable) domain
- continuous if it ranges over the real numbers

We will only consider discrete random variables with finite domains

Note: Propositional variables can be seen as random variables over the Boolean domain

Random Variables

Variable	Domain
Age	$\{1,2,...,120\}$
$Weather$	$\{sunny,dry,cloudy,rain,snow\}$
$Size$	$\{small,medium,large\}$
$Blonde$	$\{true,false\}$

The probability that a random variable X has value val is written as

$$P(X = val)$$

Note 1: $P(A = true)$ is written shortly as $P(a)$ while $P(A = false)$ is written as $P(\neg a)$

Note 2: Traditionally, in Probability Theory variables are capitalized and constant values are not

Probability Distribution

If X is a random variable, we use the **bold case** $\mathbf{P}(X)$ to denote a vector of values for the probabilities of each individual element that X can take. Example:

$$P(Weather = sunny) = 0.6$$

$$P(Weather = rain) = 0.2$$

$$P(Weather = cloudy) = 0.18$$

$$P(Weather = snow) = 0.02$$

Then $\mathbf{P}(Weather) = \mathrm{h}0.6,0.2,0.18,0.02\mathrm{i}$

(the value order of“sunny”,“rain”, cloudy”,“snow”is assumed)

$\mathbf{P}(Weather)$ is called a probability distribution for the random variable $Weather$

If $X_1,\ldots,X_n$ are random variables,

$$\mathbf{P}(X_1,\ldots,X_n)$$

denotes their joint probability distribution (JPD), an n-dimensional matrix specifying the probability of every possible combination of values for $X_1,\ldots,X_n$

Example

Sky :

	sunny	*cloudy*	*rain*	*snow*
true	0.30	0.15	0.17	0.01
false	0.30	0.05	0.01	0.01

$\{sunny,cloudy,rain,snow\}$ $Wind$: $\{true,false\}$

$\mathbf{P}(Wind,Sky)$ =

All relevant probabilities about a vector $\mathrm{h}X_1,\ldots,X_n\mathrm{i}$ of random variables can be computed from $\mathbf{P}(X_1,\ldots,X_n)$

		$S = sunny$	$S = cloudy$	$S = rain$	$S = snow$		$\mathbf{P}(W)$
W		0.30	0.15	0.17	0.01		0.63
$\neg W$		0.30	0.05	0.01	0.01		0.37
$\mathbf{P}(S)$		0.60	0.20	0.18	0.02		1.00

$P(S = rain \wedge W) = 0.17$

$P(S = rain) = P(S = rain \wedge W) + P(S = rain \wedge \neg W)$

$= 0.17 + 0.01 = 0.18$

$P(W) = 0.30 + 0.15 + 0.17 + 0.01 = 0.63$

$P(S = rain \mid W) = P(S = rain \wedge W)/P(W)$

$= 0.17/0.63 = 0.27$

A joint probability distribution $\mathbf{P}(X_1,\ldots,X_n)$ provides complete information about the probabilities of its random variables.

However, JPD's are often hard to create (again because of incomplete knowledge of the domain).

Even when available, JPD tables are very expensive, or impossible, to store because of their size.

A JPD table for n random variables, each ranging over k distinct values, has k^n entries!

A better approach is to come up with conditional probabilities as needed and compute the others from them.

An Alternative to JPD: The Bayes Rule/ Bayes Theorem

Recall that for any fact A and B,

$$P(A \wedge B) \quad = \quad P(A \mid B)P(B) \quad = \quad P(B \mid A)P(A)$$

From this we obtain the Bayes Rule:

$$P(B \mid A) \quad = \quad \frac{P(A \mid B)\, P(B)}{P(A)}$$

The rule is useful in practice because it is often easier to compute/estimate $P(A \mid B)$, $P(B)$, and $P(A)$ than to compute/estimate $P(B \mid A)$ directly

Applying the Bayes Rule

What is the probability that a patient has meningitis (M) given that he has a stiff neck (S)?

$$P(m \mid s) \quad = \quad \frac{P(s \mid m)\, P(m)}{P(s)}$$

- $P(s \mid m)$ is easier to estimate than $P(m \mid s)$ because it refers to causal knowledge: meningitis typically causes stiff neck
- $P(s \mid m)$ can be estimated from past medical cases and the knowledge about how meningitis works
- Similarly, $P(m)$ and $P(s)$ can be estimated from statistical information

The Bayes rule is helpful even in absence of (immediate) causal relationships

Conditional Independence

In terms of exponential explosion, conditional probabilities do not seem any better than JPD's for computing the probability of a fact, given $n > 1$ pieces of evidence

$$P(meningitis \mid stiffNeck \wedge nausea \wedge \cdots \wedge doubleVision)$$

However, facts do not always depend on all the evidence.

Example:

$$P(meningitis \mid stiffNeck \wedge astigmatic) = P(meningitis \mid stiffNeck)$$

$Meningitis$ and $Astigmatic$ are conditionally independent, given knowledge of $StiffNeck$

Conditional Probability Notation

Recall:

$$P(A \wedge B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

A general version holds for whole joint probability distributions, e.g.,

$$\mathbf{P}(Sky, Wind) = \mathbf{P}(Sky \mid Wind)\mathbf{P}(Wind)$$

stands for (with S for Sky and W for $Wind$):

$$P(S = sunny, W = true) = P(S = sunny \mid W = true)P(W = true)$$

$$P(S = sunny, W = false) = P(S = sunny \mid W = false)P(W = false)$$

$$\ldots$$

$$P(S = snow, W = false) = P(S = snow \mid W = false)P(W = false)$$

I.e., a 4×2 set of equations, not matrix multiplication

The chain rule is derived by successive application of product rule:

$$\mathbf{P}(X_1,\ldots,X_n)$$
$$= \mathbf{P}(X_1,\ldots,X_{n-1})\mathbf{P}(X_n|X_1,\ldots,X_{n-1})$$
$$= \mathbf{P}(X_1,\ldots,X_{n-2})\mathbf{P}(X_{n-1}|X_1,\ldots,X_{n-2})\mathbf{P}(X_n|X_1,\ldots,X_{n-1})$$
$$\ldots$$
$$= \Pi^n_{i=1}\mathbf{P}(X_i|X_1,\ldots,X_{i-1})$$

Real-Life Applications of Bayes' Theorem

Bayesian inference is very important and has found application in various activities, including medicine, science, philosophy, engineering, sports, law, etc. Some of the most common applications of Bayes' Theorem in real life are :

- Spam Filtering
- Weather Forecasting
- DNA Testing
- Financial Forecasting
- Fault Diagnosis in Engineering
- Drug Testing

Let's discuss these applications in detail.

Applications of Bayes' Theorem in Spam Filtering

Spam filters in email systems use Bayes' Theorem to distinguish between legitimate emails and spam. By analyzing the frequency of certain words or phrases in both spam and non-spam emails, the filter can assign probabilities to incoming messages being spam.

- **For example,** if an email contains words commonly found in spam messages, such as "free" or "discount," the filter will calculate the likelihood that it is spam and then route it accordingly.

Applications of Bayes' Theorem in Weather Forecasting

Meteorologists apply Bayes' Theorem to improve the accuracy of weather forecasts. By incorporating data from various sources such as satellite imagery, weather stations, and historical patterns, forecast models can calculate the probability of different weather outcomes.

- **For instance,** if there is a high pressure system moving in from the west, Bayes' Theorem helps meteorologists assess the likelihood of clear skies versus rain in a particular region.

Applications of Bayes' Theorem in DNA Testing

In forensic science, Bayes' Theorem is used to interpret DNA evidence in criminal investigations. By comparing DNA samples collected from crime scenes to databases of known DNA profiles, analysts can calculate the probability that a suspect's DNA matches the evidence.

- **For example,** if the DNA found at the crime scene matches that of the suspect, Bayes' Theorem helps quantify the strength of that evidence in court.

Applications of Bayes' Theorem in Financial Forecasting

Bayes' Theorem is applied in finance to assess the risk and return of investment portfolios. By analyzing historical market data and economic indicators, financial analysts can calculate the probability of different investment outcomes.

- **For example,** if a portfolio is heavily weighted towards technology stocks, Bayes' Theorem helps investors estimate the likelihood of achieving a certain rate of return based on past performance and market trends.

Applications of Bayes' Theorem in Drug Testing

In pharmacology, Bayes' Theorem is used to evaluate the effectiveness of new drugs in clinical trials. By comparing patient outcomes between treatment and control groups, researchers can calculate the probability that a drug is effective in treating a particular condition.

- **For example,** if patients receiving the drug show a significant improvement in symptoms compared to those receiving a placebo, Bayes' Theorem helps quantify the likelihood that the drug is beneficial.

Bayesian networks:

- A data structure to represent the dependencies among variables and to give a concise specification of any full joint probability distribution.
- Also called belief networks or probabilistic network or casual network or knowledge map.

The basic idea is:

- Knowledge in the world is modular -- most events are conditionally independent of most other events.
- Adopt a model that can use a more local representation to allow interactions between events that only affect each other.
- Some events may only be unidirectional others may be bidirectional -- make a distinction between these in model.
- Events may be causal and thus get chained together in a network.

A Bayesian network is a directed acyclic graph which consists of:

- A set of random variables which makes up the nodes of the network.
- A set of directed links (arrows) connecting pairs of nodes. If there is an arrow from node X to node Y, X is said to be a parent of Y.
- Each node Xi has a conditional probability distribution P(Xi| Parents(Xi)) that quantifies the effect of the parents on the node.

Intuitions:

- A Bayesian network models our incomplete understanding of the causal relationships from an application domain.
- A node represents some state of affairs or event.
- A link from X to Y means that X has a direct influence on Y.

Implementation

- A Bayesian Network is a directed acyclic graph:
  - o A graph where the directions are links which indicate dependencies that exist between nodes.
  - o Nodes represent propositions about events or events themselves.
  - o Conditional probabilities quantify the strength of dependencies.

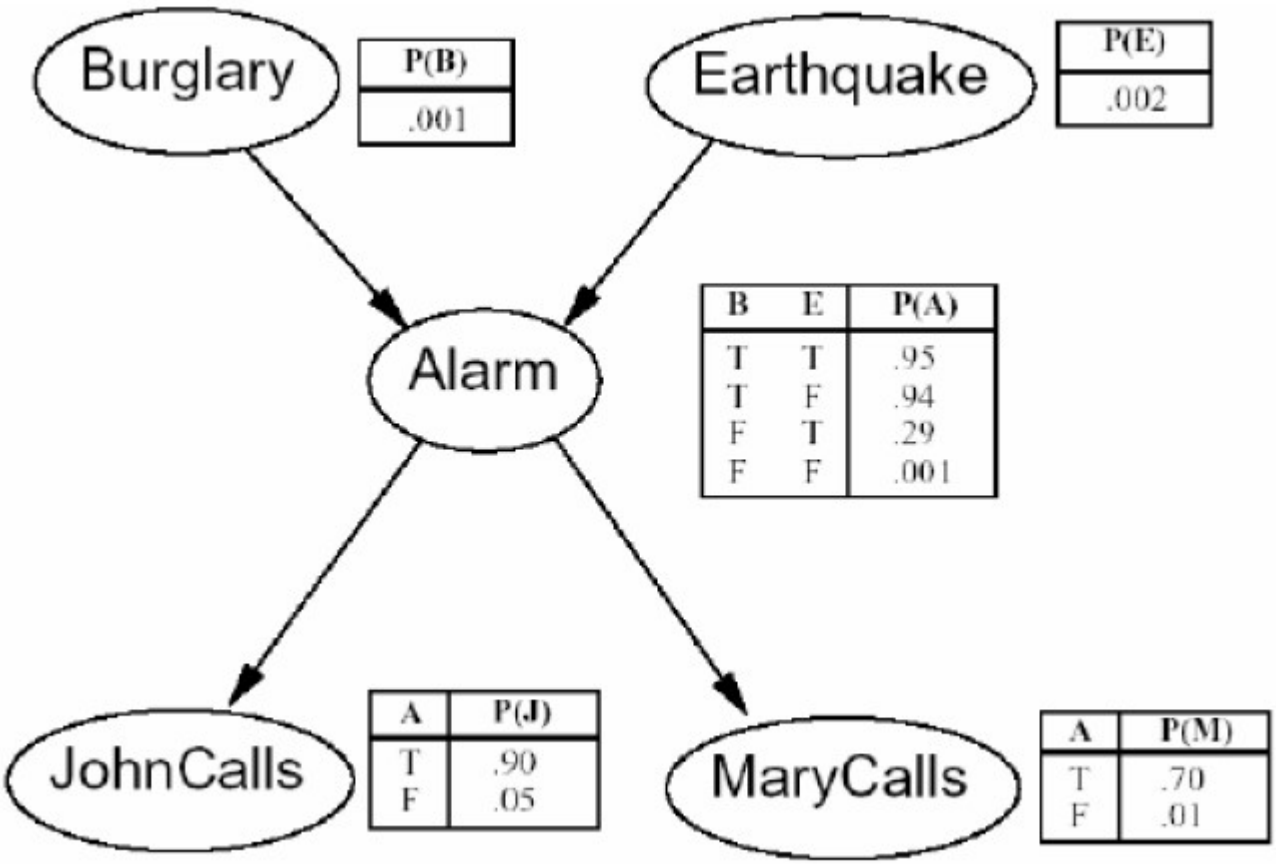
Example:

Sample Domain:

You have a burglar alarm installed in your home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and sometimes misses the alarm altogether.

We would like have to estimate the probability of a burglary with given evidence who has or has not call.

Variables:Burglary, Earthquake, Alarm, JohnCalls, MaryCalls



P(B)
.001

P(E)
.002

B	E	P(A)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J)
T	.90
F	.05

A	P(M)
T	.70
F	.01

The probabilities associated with the nodes reflect our representation of the causal relationships.

A Bayesian network provides a complete description of the domain in the sense that one can compute the probability of any state of the world (represented as a particular assignment to each variable).

Example: What is the probability that the alarm has sounded, but neither burglary nor an earthquake has occurred, and both John and Mary call?

$P(j, m, a, \neg b, \neg e) = P(j|a)\ P(m|a)\ P(a|, \neg b, \neg e)\ P(\neg b)\ P(\neg e)$

$= 0.90*0.70*0.001*0.999*0.998 = 0.00062$